CSCI 580 - Assignment #4

This report uses the results saved in my Google Colab notebook (40 cities, city seed 7).

1. Problem and Goal

The Traveling Salesman Problem (TSP) is to visit every city once and return to the starting city with the shortest total route. It gets difficult when there are many cities because there are many possible routes. I tested two methods from class, Simulated Annealing (SA) and a Genetic Algorithm (GA), and compared them with a Nearest Neighbor (NN) starting route.

2. How I Implemented the Methods

Nearest Neighbor baseline

The NN method starts at city 0 and repeatedly picks the closest city that has not been visited. It gives a quick route to compare against, but it does not always find the shortest route.

Simulated Annealing (SA)

SA starts from the NN tour. Every iteration, it tries a 2-opt change by reversing part of the route. Shorter routes are accepted. A longer route can also be accepted sometimes, based on the current temperature. The temperature goes down each iteration, so longer routes become less likely to be accepted.

```
accept = delta <= 0 or rng.random() < math.exp(-delta / max(T, 1e-12))
```

```
T = max(T * cfg.alpha, 1e-12)
```

Genetic Algorithm (GA)

GA works with a population of tours. Tournament selection picks a better parent from a small random group. Order crossover combines two parents without repeating cities, and swap mutation switches two cities. Elitism keeps the best few tours for the next generation.

```
winner = min(choices, key=lambda i: lengths[i])
```

```
if rng.random() < cfg.mutation_rate: child = mutate_swap(child, rng)
```

3. Experiment Setup

Setting	SA	GA
Cities / random seed	40 / 7	40 / 7
Iterations / generations	20,000 iterations	400 generations
Starting temperature / alpha	0.2 / 0.9995	N/A
Population	N/A	200
Tournament size	N/A	5
Crossover / mutation	2-opt moves	0.9 / 0.2
Elites	N/A	5
Algorithm seed	123	999

4. Results

Method	Best tour length	Improvement vs NN
Nearest Neighbor	6.0214	-
SA	5.3626	10.94%
GA	5.3672	10.86%

Both SA and GA improved on NN by more than 10%. SA got the shortest route in this run, but its result was only about 0.0046 shorter than GA.

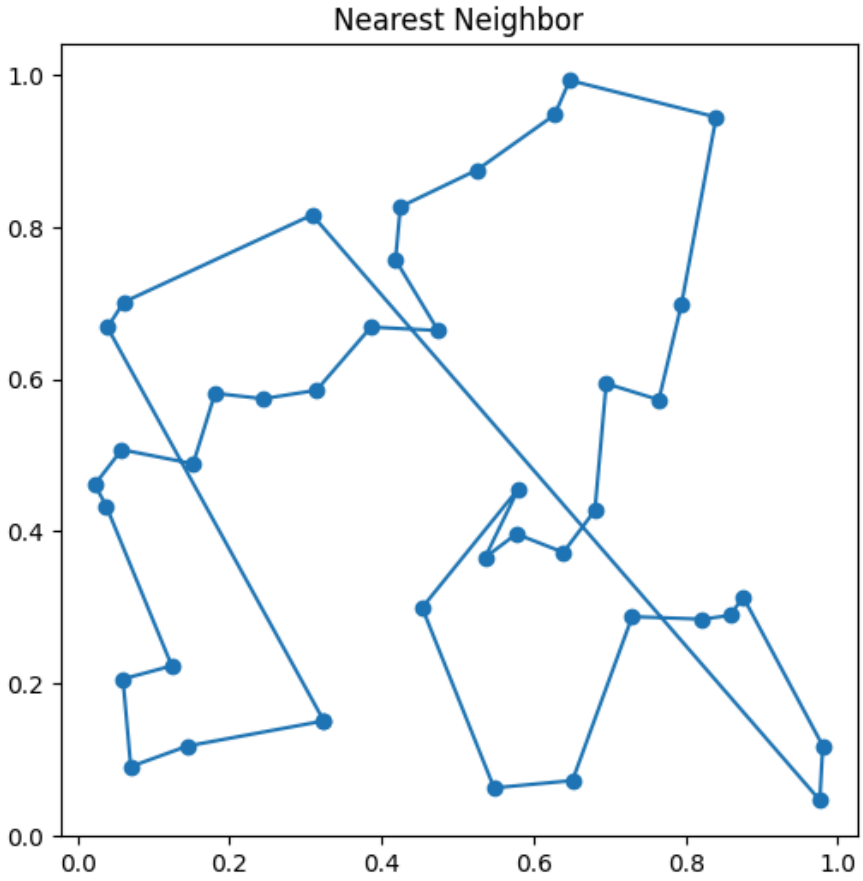


Figure 1. Nearest Neighbor starting tour.

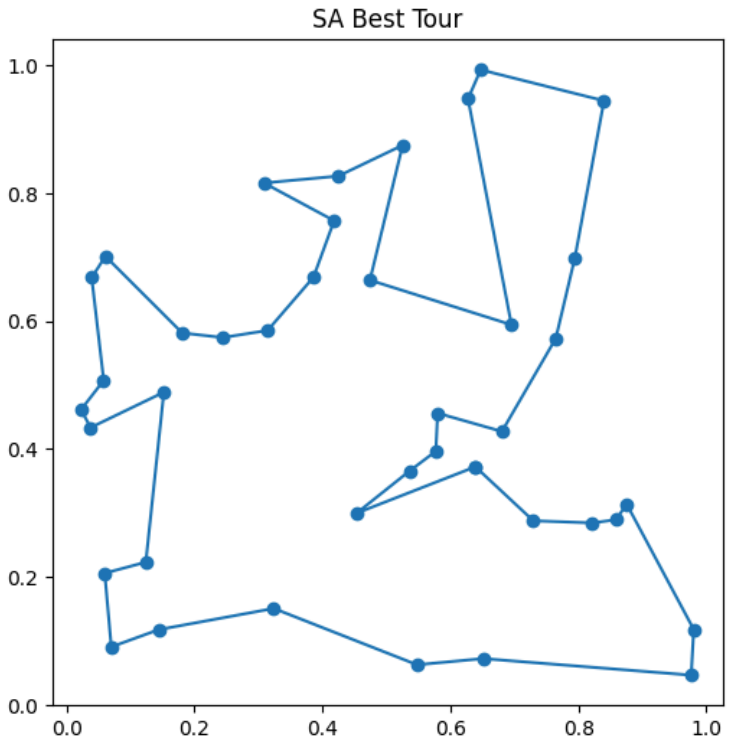


Figure 2. SA best tour.

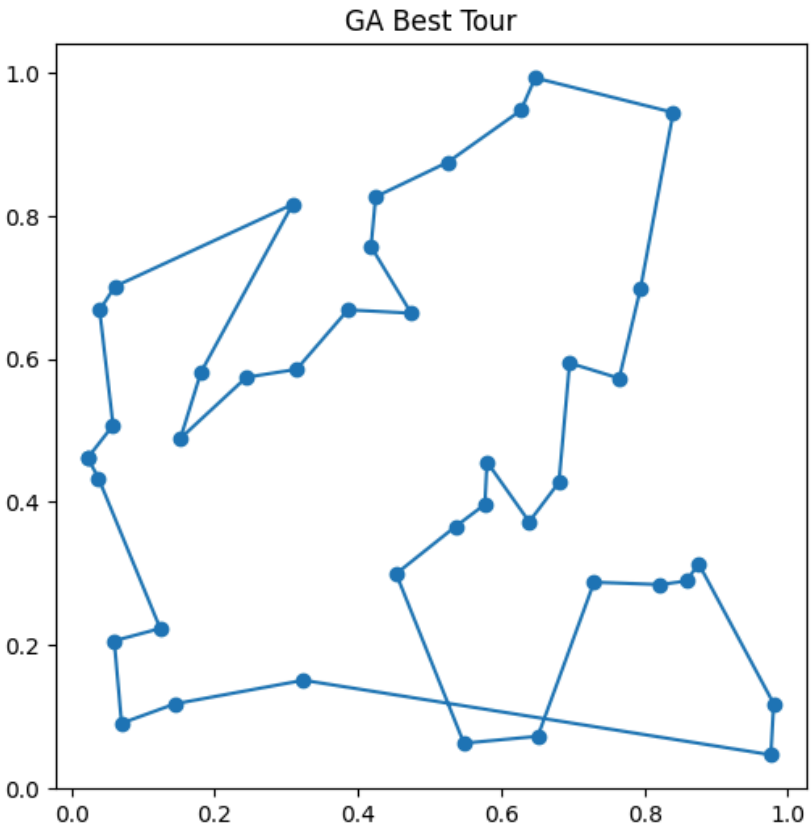


Figure 3. GA best tour.

5. Part 6 - Data and Discussion

Which algorithm converged faster?

GA found its final reported best length by generation 50 (the first progress printout), while SA reached its final reported best by iteration 8,000. The plot with tour evaluations gives a fairer way to compare their progress than comparing a generation directly with an iteration. On these saved runs, SA and GA both improved early and then stopped finding better routes. Since GA evaluates about 200 tours per generation, raw generation and iteration numbers are not directly comparable. The number of evaluations plotted for GA is an approximation because it counts a whole population per generation.

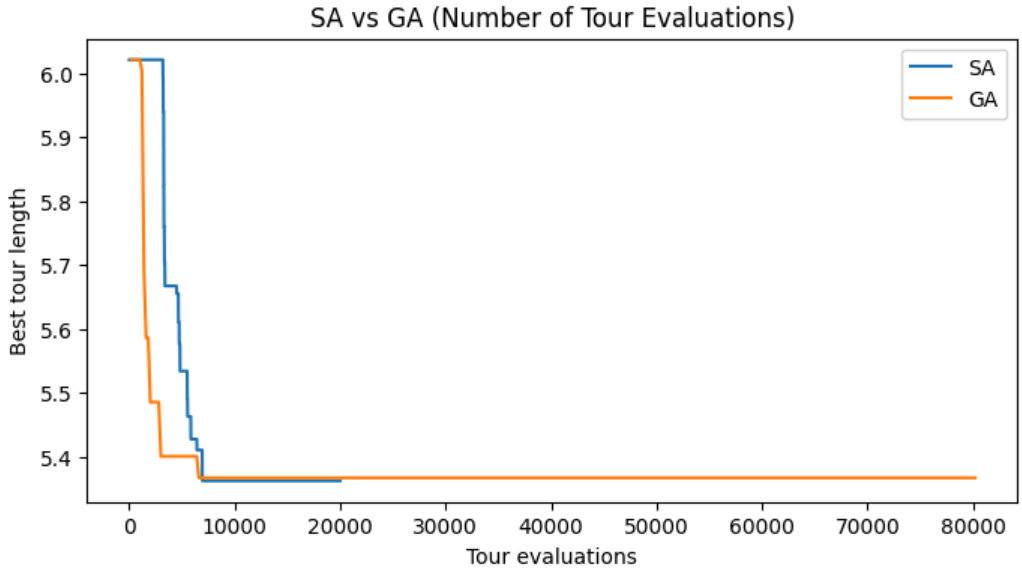


Figure 4. SA and GA best length versus approximate tour evaluations.

Which showed more variability?

The saved notebook only shows one seeded run of each algorithm, so it does not measure variability between runs. SA changes one current solution at a time, while GA works with many routes at once. To say which method has more run-to-run variability, I would need to repeat both methods with different random seeds and compare their results. In this run, the GA population also had different route lengths, as shown by its average and best curves.

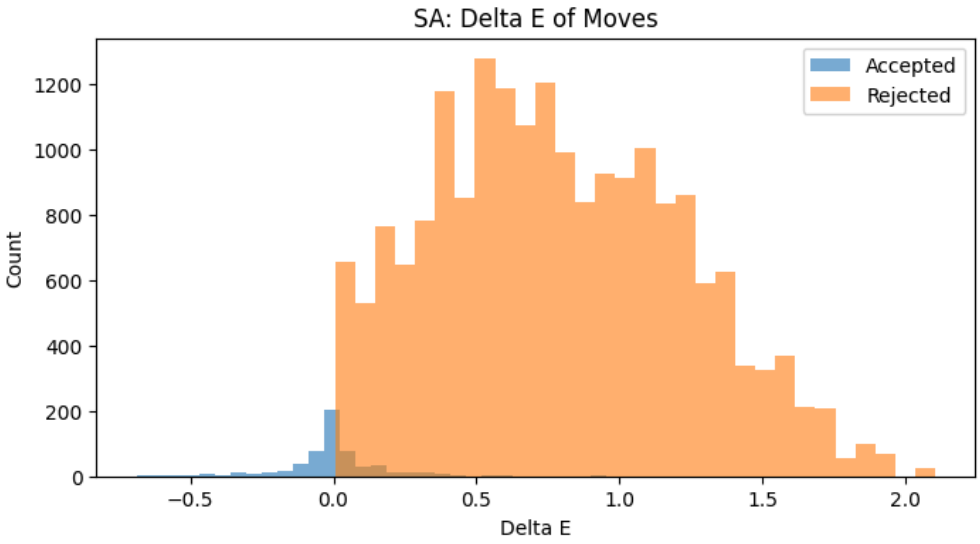


Figure 5. SA accepted and rejected move changes (delta E).

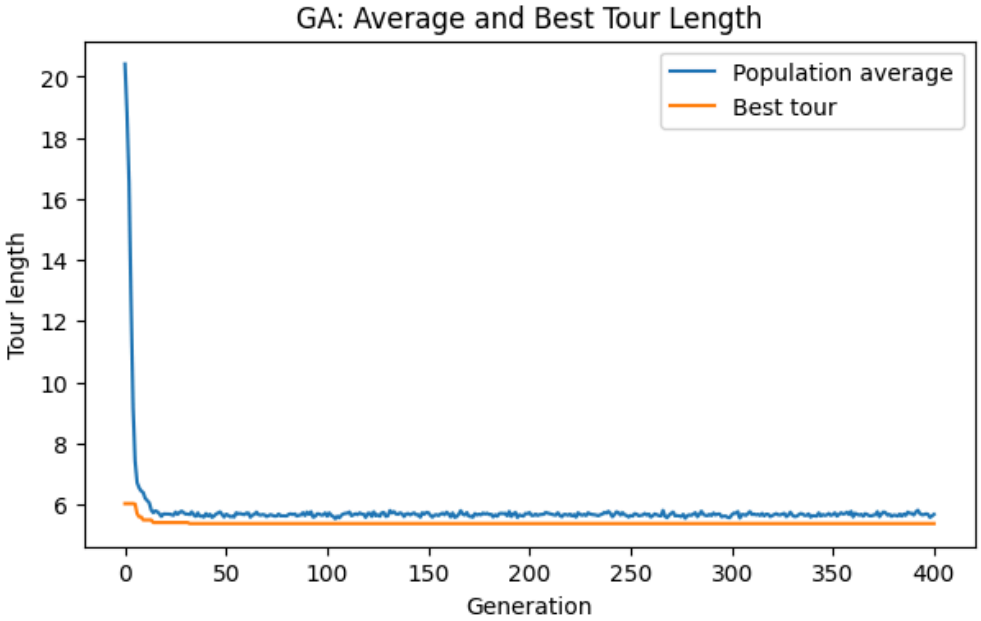


Figure 6. GA average and best tour length.

Did SA escape local minima? What is the evidence?

SA can accept a move that makes the route longer, which helps it keep exploring instead of accepting only improvements. The code uses exp(-delta / T) for a positive delta. The accepted and rejected delta E histogram is evidence of the types of moves that SA took. This supports that SA can escape a local minimum; however, the saved output does not directly prove that a specific local minimum was escaped. For that, I would track an accepted positive-delta move followed by a new best solution.

Did GA lose diversity? What is the evidence?

Yes, the saved data shows lower diversity by the end. The fraction of unique tours in the population dropped from 1.00 at the start to 0.18 at the end. That means many tours became copies of other tours. The best

solution also stayed at 5.3672 after the first 50-generation report, which suggests the population had stopped making progress in this run.

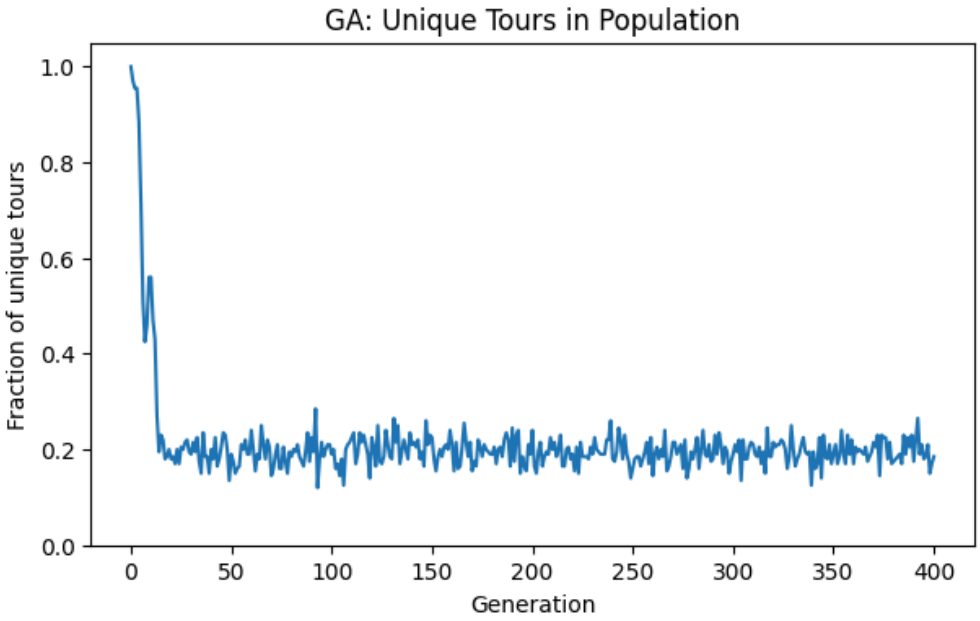


Figure 7. Fraction of unique GA tours in the population.

6. Conclusion

Both methods gave better routes than NN. SA was slightly better in this test, with 5.3626 compared with 5.3672 for GA. SA slowly lowers the chance of accepting worse moves, while GA searches with a population. One issue I saw with GA is that the population became less diverse. These are results for this one TSP instance and random seed, so I cannot say SA is always better.